

# Structural Search Geometry

A MAAT v1.6 Framework Note with a Toy Search Benchmark

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## Abstract

MAAT v1.4 introduced multi-scale defect fields, and MAAT v1.5 added the requirement that supports be defined relative to the generator of the system. This note formulates the next step, MAAT v1.6, as a structural search geometry: structural supports are used as priority-modulating coordinates along a search trajectory through a complex candidate space.

The central claim is deliberately narrow. Structural selection should not replace verification, objective functions, or domain-specific heuristics. Instead, it can act as a search-geometry layer that modulates where effort is spent when a search space contains traps, bridges, fragile basins, or translation bottlenecks.

We test this idea in a synthetic graph-search benchmark with smooth, trap, bridge, and trap+bridge landscapes. The result is mixed and useful. An anchored MAAT v1.6 score reaches all targets and improves over energy-only search in trap-like cases, but simple distance and connectivity baselines remain faster on these clean grid tasks. A structure-heavy variant is substantially worse. The benchmark therefore supports a design rule rather than a victory claim: *structural guidance should modulate verification-driven search, not replace the target or proof obligation.*

## 1 Purpose and Status

Earlier MAAT papers treated structural selection mainly as a diagnostic, ranking, or stability layer. The recent mathematical-discovery note reframed the same idea as a possible ranking layer for proof search and counterexample search. MAAT v1.6 makes this transition explicit.

The object is no longer only a state  $X$ . It is a search trajectory through a candidate space:

$$X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_T, \quad X_t \in \mathcal{X}.$$

The guiding question becomes:

*Which structures are reachable, stabilised, prioritised, or discovered under constrained search?*

This note is a framework and toy-benchmark paper. It does not claim that MAAT v1.6 is an optimal search algorithm. It tests whether the search-geometry interpretation is coherent enough to produce a reproducible benchmark with meaningful failure modes.

## 2 From State Selection to Search Geometry

Let  $\mathcal{X}$  be a search space. It may be a space of field configurations, SAT assignments, graph constructions, model parameters, mathematical candidates, hypotheses, or representations. Let

$$\mathcal{G}_{\text{search}} : \mathcal{X} \rightarrow \mathcal{X}$$

denote a search generator or transition rule. Examples include a solver step, a proof-search expansion, a graph rewrite, a local optimisation move, a mutation operator, or a scientific hypothesis update.

The v1.6 view is:

supports act as priority-modulating coordinates on a search trajectory.

They do not determine truth. They influence search priority.

For a candidate  $x \in \mathcal{X}$ , define local supports

$$\Gamma_a(x) = \frac{1}{1 + d_a(x)}, \quad a \in \{H, B, S, V\},$$

and the v1.2.1 robustness closure

$$R_{\text{resp}}(x) = (H(x)B(x)V(x))^{1/3}, \quad (1)$$

$$R_{\text{rob}}(x) = \min\left(R_{\text{resp}}(x), (H(x)B(x)S(x)V(x))^{1/4}\right). \quad (2)$$

The v1.6 search score adds translation and verification terms:

$$F_{\text{search}}(x) = \lambda_D d_{\text{ver}}(x) + \lambda_E E(x) + \lambda_M \left[ - \sum_a \log(\varepsilon + \Gamma_a(x)) \right] + \lambda_R(1 - R_{\text{rob}}(x)) - \lambda_V V(x). \quad (3)$$

Here  $d_{\text{ver}}$  measures distance to the verification target or success condition,  $E$  is a domain objective or energy proxy, and  $V$  rewards connective search positions such as bridges, translation points, or high-flow regions.

Operationally, the “geometry” is the priority landscape induced by  $F_{\text{search}}$  over the directed frontier generated by  $\mathcal{G}_{\text{search}}$ . Equivalently, the pair  $(\mathcal{G}_{\text{search}}, F_{\text{search}})$  turns a candidate space into a search-biased graph: local supports do not change which candidates are valid, but they change which candidates are expanded first. Equation (3) is intentionally not universal in its weights. If the structural term dominates the verification term, search can become coherent but irrelevant.

### 3 Toy Benchmark

We test the idea on synthetic graph-search spaces. Each instance is a  $34 \times 34$  grid graph with a fixed start and target. Four environment families are generated:

- **smooth**: no obstacle or trap structure,
- **trap**: an attractive decoy basin in the energy landscape,
- **bridge**: a wall with a narrow off-axis passage,
- **trap+bridge**: both a decoy basin and a narrow passage.

For each family we generate 14 instances. The search budget is 850 node expansions. The comparison methods are:

| Method             | Meaning                                                      |
|--------------------|--------------------------------------------------------------|
| random frontier    | random expansion from the frontier.                          |
| distance-only      | best-first search using Euclidean target distance.           |
| energy-only        | best-first search using the trap-distorted energy landscape. |
| connectivity-only  | distance plus a simple connectivity/bridge correction.       |
| MAAT v1.6 anchored | verification-dominant MAAT structural guidance.              |
| MAAT v1.6 heavy    | structure-heavy MAAT guidance.                               |

The two MAAT variants are deliberately included to test the failure mode. The anchored version keeps  $d_{\text{ver}}$  dominant. The heavy version gives structural support much more influence.

## 4 Operational Supports

The toy implementation uses simple local graph quantities. They are not claimed to be fundamental definitions. They are first operational coordinates for the search-geometry idea.

- $H$  rewards local coherence by penalising fragile attractive wells and high local energy curvature.
- $B$  rewards balanced exits from a local state rather than one-sided dead-end behaviour.
- $S$  rewards controlled useful activity, i.e. moderate local improvement rather than maximum jumpiness.
- $V$  rewards connectivity and bridge-like search positions using local degree and approximate betweenness centrality.
- $R_{\text{rob}}$  is the v1.2.1 emergent robustness closure.

Thus the benchmark tests whether these structural coordinates help navigation in a search space with traps and bridges.

## 5 Results

Table 1: Aggregate results over all four search families. Lower expansions are better.

| Algorithm          | Success rate | Median expansions |
|--------------------|--------------|-------------------|
| distance-only      | 1.000        | 59.0              |
| connectivity-only  | 1.000        | 59.0              |
| MAAT v1.6 anchored | 1.000        | 92.0              |
| energy-only        | 1.000        | 106.0             |
| MAAT v1.6 heavy    | 1.000        | 234.0             |
| random frontier    | 0.741        | 702.5             |

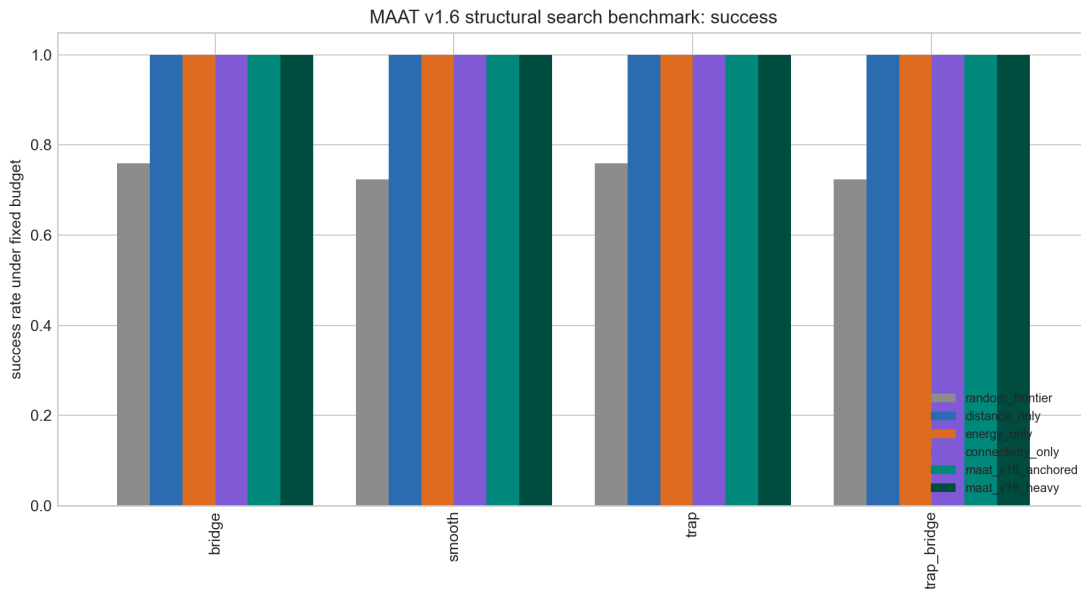


Figure 1: Success rates under a fixed expansion budget. All deterministic guided searches reach all targets, while random frontier search fails on a substantial fraction of runs.

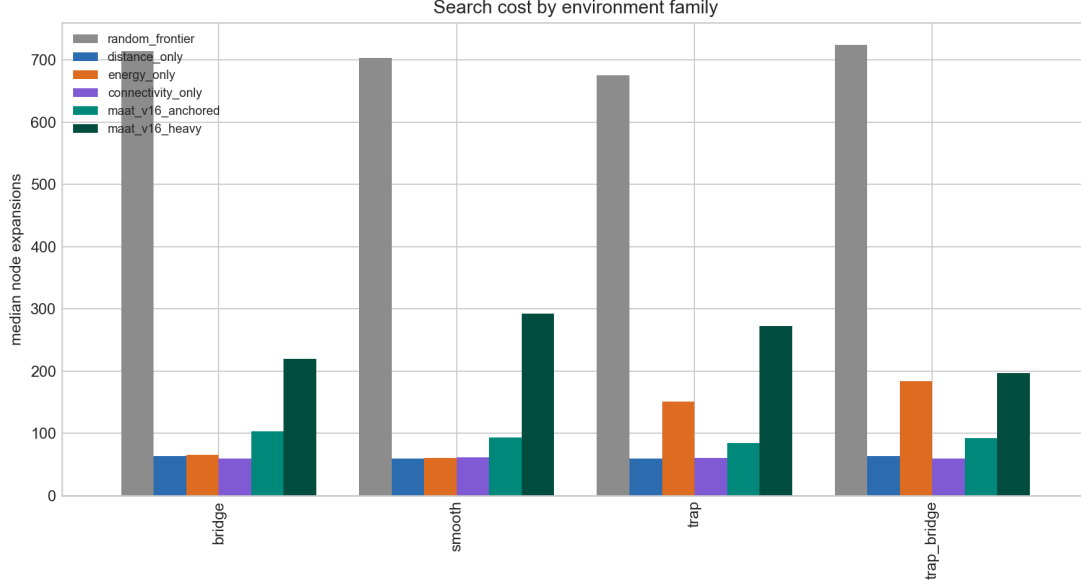


Figure 2: Median node expansions. Simple distance and connectivity baselines dominate these clean grid tasks. The anchored v1.6 score remains functional and beats energy-only in trap-like cases, while the structure-heavy variant is too slow.

The trap-like cases show the intended partial signal. Relative to energy-only search, the anchored v1.6 score reduces median expansions by about 44% in the trap family and about 50% in the trap+bridge family:

Table 2: Trap-family comparison. Lower expansions are better.

| Family      | MAAT v1.6 anchored | Energy-only |
|-------------|--------------------|-------------|
| trap        | 84.5               | 150.5       |
| trap+bridge | 92.0               | 183.5       |

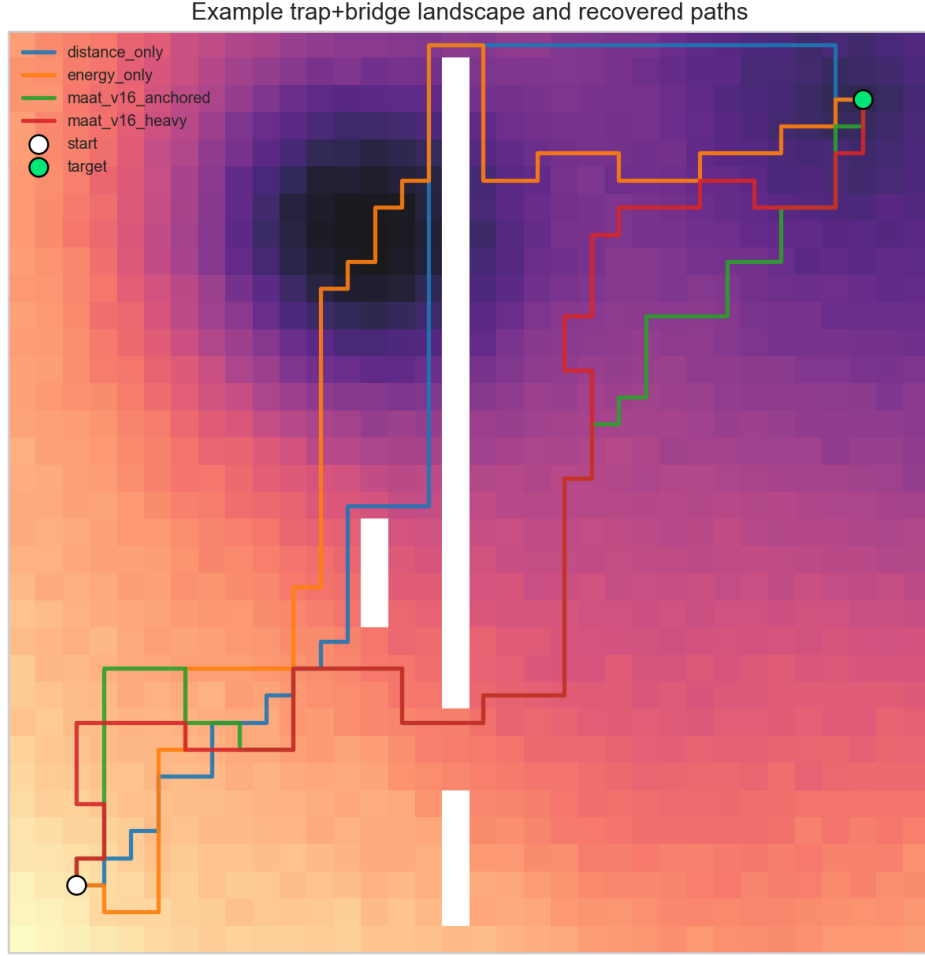


Figure 3: Example trap+bridge landscape. The paths illustrate how different priority rules traverse the same synthetic energy landscape.

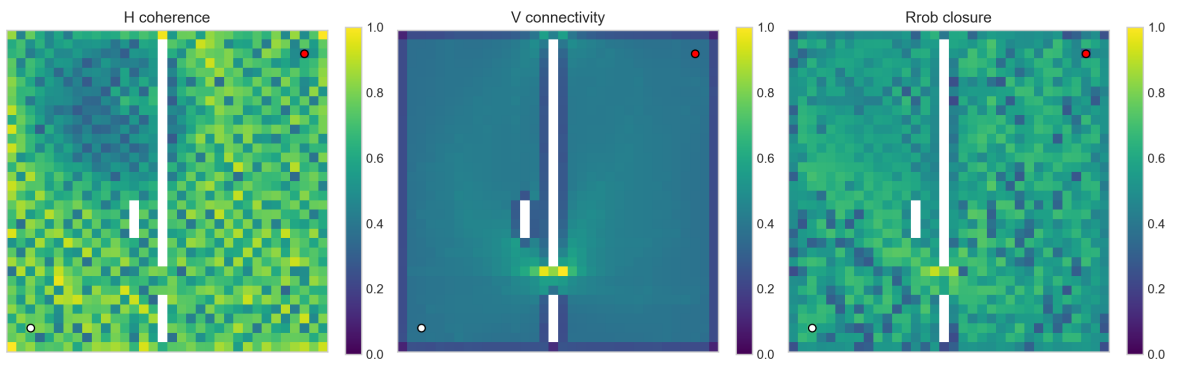


Figure 4: Structural support maps on the same search space. The supports act as navigation fields rather than truth labels.

## 6 Interpretation

The benchmark gives neither a triumph nor a failure. It establishes three useful lessons.

First, search geometry must remain anchored. The structure-heavy MAAT variant is much slower than the anchored variant. This means that structural support is not enough: a search

process still needs a verification target, objective, proof obligation, or success condition.

Second, structural guidance can regularise misleading energy. In the trap and trap+bridge families, the anchored v1.6 score improves over energy-only search. This is the positive signal in the benchmark: it does not show superiority over simple geometry, but it shows that structural coordinates can partially protect a search from attractive but misleading objective basins.

Third, simple baselines remain strong. Distance-only and connectivity-only dominate the clean grid tasks. This is an important negative result. The present v1.6 implementation does not outperform simple geometric heuristics on simple synthetic spaces. Its value, if any, should be sought in richer spaces where the target distance is not directly available, local evaluations are costly or noisy, bridges are not simple graph bottlenecks, and verification requires expensive downstream checks.

## 7 Relation to MAAT v1.4 and v1.5

The result fits the recent MAAT development. MAAT v1.4 showed that global scalar compression can lose local defect geometry. MAAT v1.5 showed that support coordinates must be defined relative to a generator. MAAT v1.6 adds the search perspective:

defect fields become navigation fields on structured search spaces.

The benchmark also clarifies a boundary:

structural navigation is not a substitute for verification.

This boundary is important for mathematical discovery, SAT, AI search, scientific hypothesis ranking, and model-space exploration.

## 8 Limitations

- The benchmark is synthetic, low-dimensional, and geometrically clean. Because the target is visible and Euclidean distance is cheap, the dominance of distance-only and connectivity-only baselines is expected rather than surprising.
- The supports are heuristic operational definitions, not derived universally.
- Distance and connectivity baselines dominate the tested clean grid spaces.
- No claim is made about optimal search, theorem proving, SAT solving, or machine-learning training.
- The next test should use harder external search spaces where local traps, bridge states, and verification bottlenecks are not geometrically trivial, for example theorem-search graphs, CDCL/SAT state spaces, graph construction problems, model-configuration spaces, or optimisation tasks with expensive validation.

## 9 Reproducibility

The experiment is stored in:

`experiments/structural_search_geometry_v16/`

Run:

```
cd experiments/structural_search_geometry_v16
python3 structural_search_geometry_v16.py
```

Outputs are written to:

outputs\_v16/

The public repository is:

<https://github.com/Chris4081/structural-selection-principle>

## 10 Conclusion

MAAT v1.6 reframes structural selection as search geometry. The central idea is that structural supports can guide navigation through complex candidate spaces, including mathematical, computational, physical, and scientific search spaces.

The first toy test gives a conservative result. Anchored structural search is functional and improves over energy-only search in trap-like landscapes, but it does not beat simple distance or connectivity baselines on clean grid tasks. Heavy structural weighting is actively harmful.

Thus the operational v1.6 design rule is:

*Use structural supports to modulate verification-driven search. Do not let structural support replace the target, proof obligation, or validation criterion.*

This is a useful constraint on future MAAT search applications. It turns v1.6 into a testable search-geometry programme rather than a broad claim about universal discovery.

## References

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- [3] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, Pearson, 4th edition (2021).
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